

MODULE 7: TESTING AND DEBUGGING

Project: Unified Military Analytics and Comparison Dashboard

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Domain: Data Visualization

Tool Used: Power BI

1. Methodology

Module 7 focused on validating the functional accuracy, performance stability, and visual integrity of the Unified Military Analytics and Comparison Dashboard developed in Power BI. The objective of this phase was to ensure that all dashboards operate correctly under various user interactions and that all displayed values are accurate and consistent with the source dataset.

The testing process was conducted in a structured manner covering data validation, slicer functionality, DAX measure accuracy, navigation flow, layout consistency, performance optimization, and edge-case scenario handling.

A. Data Accuracy Verification

All numerical values displayed across dashboards were manually verified against the original dataset. Random spot checks were performed on multiple countries to confirm the correctness of:

- Manpower
- Aircraft count
- Naval fleet strength
- Defense budget
- Power Index rank
- Assets per capita

Coalition totals were manually validated by summing selected countries and comparing them with dynamically aggregated DAX results. This ensured that aggregation logic using SUM and CALCULATE functions was functioning correctly.

B. Slicer and Filter Testing

All slicers were tested individually and in combination to verify correct filter propagation. This included:

- Country slicers in Quick Stats and Nation Overview
- Country A and Country B slicers in Compare Powers
- Multi-select coalition slicer in Coalition Builder
- Region, Continent, and Alliance filters

Each slicer was tested under single selection, multi-selection, and cleared-selection scenarios to confirm accurate visual updates and proper cross-filter behavior. Sync Slicers functionality was verified to ensure consistent filtering across report pages.

C. DAX Measure Validation

All calculated measures were reviewed to ensure logical correctness and accurate context handling. Special attention was given to:

- Country comparison logic using CALCULATE
- Coalition aggregation measures
- Handling of no-selection or same-country selection scenarios
- Context transition behavior across visuals

Measures were tested for stability and correctness across different dashboard interactions to prevent incorrect totals or blank outputs.

D. Navigation and Integration Testing

Page navigation buttons were tested to ensure seamless movement between the following dashboards:

- Quick Stats
- Nation Overview
- Compare Powers
- Coalition Builder

Navigation logic was validated to confirm that filter context was preserved where required and that no broken links or misdirected pages existed. Full dashboard integration was verified to ensure smooth user experience across the entire report.

E. Visual and Layout Debugging

All visuals were reviewed to eliminate layout and formatting inconsistencies. The following were validated:

- Correct chart titles and labels
- Consistent number formatting (Thousands, Millions, Billions)
- No overlapping visuals
- Proper alignment and spacing
- Uniform theme and typography
- Absence of unnecessary scrollbars

This ensured professional presentation quality and usability.

F. Performance Optimization

Performance testing was conducted to confirm that:

- The report loads efficiently
- Slicers respond without delay

- Visuals render smoothly
- No redundant DAX measures exist
- Relationships follow a proper one-to-many structure
- Star-schema model structure is maintained

The data model was reviewed and cleaned to remove unnecessary columns and optimize report size.

G. Edge Case and Scenario Testing

Edge cases were tested to ensure system stability under extreme or unusual selections, including:

- Highest-ranked country selection
- Lowest-ranked country selection
- Same country selected in Compare Powers
- No country selected
- Coalition with single country
- Coalition with multiple countries

This ensured that the dashboard behaves predictably and reliably in all user scenarios.

2. QA Checklist

A structured QA checklist was created and executed to validate all functional, analytical, and visual components of the Power BI report. The checklist covered data validation, slicer behavior, DAX logic, navigation, layout quality, performance stability, and edge-case testing.

All checklist items were reviewed and confirmed prior to final submission.

3. Deliverables Achieved

- Debugged and fully polished Power BI workbook
- Completed QA checklist documentation
- Validated DAX calculations and aggregation logic
- Verified navigation and slicer synchronization
- Optimized and performance-tested report
- Final file saved as: `global_military_firepower_2025.pbix`

Module 7 Outcome

Module 7 successfully ensured the reliability, accuracy, and performance quality of the Unified Military Analytics and Comparison Dashboard. Through systematic testing and debugging, all dashboards were validated for correctness, usability, and analytical consistency. The final Power BI report is fully functional, integrated, and ready for professional submission.